

Chapter 2: Matrices

All Examples with Step-by-Step Solutions

Based on Ron Larson, Elementary Linear Algebra, 8th edition

Prepared as a study PDF with formulas and working-out steps placed on separate lines for readability.

How to use this PDF: Work through one example at a time. Cover the final answer, try the next calculation yourself, then compare your step with the worked solution.

Notation: O means a zero matrix. I_n means the identity matrix of order n . A^{-1} means the inverse of A . A^T means the transpose of A .

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2.1 Operations with Matrices

Core skills in this section: identify matrix size, compare matrices, add and subtract matrices, multiply by a scalar, multiply matrices, write systems as $Ax = b$, and interpret Ax as a linear combination of the columns of A .

Example 2.1.1 - Equality of Matrices

Decide when the matrices are equal.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 3 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 2 \\ x & 4 \end{bmatrix}$$

Step 1: Compare A and B by size.

A is 2×2 .
B is 2×1 .
The sizes are different.
Therefore, $A \neq B$.

Step 2: Compare B and C by size.

B is 2×1 .
C is 1×2 .
The sizes are different.
Therefore, $B \neq C$.

Step 3: Compare A and D. They are both 2×2 , so compare matching entries.

A = D requires:
 $1 = 1$
 $2 = 2$
 $3 = x$
 $4 = 4$
So, $x = 3$.
Therefore, $A = D$ only when $x = 3$.

Example 2.1.2 - Addition of Matrices

Part (a)

$$\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$$

Add corresponding entries.

$$\begin{aligned} &= \begin{bmatrix} -1 + 1 & 2 + 3 \\ 0 + (-1) & 1 + 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix} \end{aligned}$$

Part (b)

$$\begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Adding a zero matrix does not change the matrix.

$$\begin{bmatrix} 0 & 1 & -2 \\ 1 & 2 & 3 \end{bmatrix}$$

Part (c)

$$\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + (-1) \\ -3 + 3 \\ -2 + 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Part (d)

$$\begin{bmatrix} 2 & 1 \\ 4 & 0 \\ 1 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$$

First matrix: 3×2

Second matrix: 2×2

Sizes are different.

Therefore, the sum is undefined.

Example 2.1.3 - Scalar Multiplication and Subtraction

Given:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix}$$

Part (a): Find $3A$

$$\begin{aligned} 3A &= 3 \begin{bmatrix} 1 & 2 & 4 \\ -3 & 0 & -1 \\ 2 & 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) & 3(2) & 3(4) \\ 3(-3) & 3(0) & 3(-1) \\ 3(2) & 3(1) & 3(2) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} \end{aligned}$$

Part (b): Find $-B$

$$\begin{aligned} -B &= (-1)B \\ -B &= \begin{bmatrix} -2 & 0 & 0 \\ -1 & 4 & -3 \\ 1 & -3 & -2 \end{bmatrix} \end{aligned}$$

Part (c): Find $3A - B$

$$\begin{aligned} 3A - B &= \begin{bmatrix} 3 & 6 & 12 \\ -9 & 0 & -3 \\ 6 & 3 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 1 & -4 & 3 \\ -1 & 3 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 - 2 & 6 - 0 & 12 - 0 \\ -9 - 1 & 0 - (-4) & -3 - 3 \\ 6 - (-1) & 3 - 3 & 6 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 6 & 12 \\ -10 & 4 & -6 \\ 7 & 0 & 4 \end{bmatrix} \end{aligned}$$

Example 2.1.4 - Concession Sales Using Matrix Multiplication

A stadium sells peanuts, hot dogs, and sodas at three stands. Quantities sold are in the matrix Q , and prices are in the price vector p .

$$Q = \begin{bmatrix} 120 & 250 & 305 \\ 207 & 140 & 419 \\ 29 & 120 & 190 \end{bmatrix} \quad p = \begin{bmatrix} 2.00 \\ 3.00 \\ 2.75 \end{bmatrix}$$

The product Qp gives total sales at each stand.

$$Qp = \begin{bmatrix} 120(2.00) + 250(3.00) + 305(2.75) \\ 207(2.00) + 140(3.00) + 419(2.75) \\ 29(2.00) + 120(3.00) + 190(2.75) \end{bmatrix}$$

South stand:

$$\begin{aligned} &120(2.00) + 250(3.00) + 305(2.75) \\ &= 240 + 750 + 838.75 \\ &= 1828.75 \end{aligned}$$

North stand:

$$\begin{aligned} &207(2.00) + 140(3.00) + 419(2.75) \\ &= 414 + 420 + 1152.25 \\ &= 1986.25 \end{aligned}$$

West stand:

$$\begin{aligned} &29(2.00) + 120(3.00) + 190(2.75) \\ &= 58 + 360 + 522.50 \\ &= 940.50 \end{aligned}$$

$$Qp = \begin{bmatrix} 1828.75 \\ 1986.25 \\ 940.50 \end{bmatrix}$$

Example 2.1.5 - Product of Two Matrices

Find AB .

$$A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \quad B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$$

Check sizes first.

A is 3×2 .

B is 2×2 .

The inner sizes match, so AB is defined.

The product AB will be 3×2 .

Compute each entry by row times column.

$$\begin{aligned} c_{11} &= (-1)(-3) + 3(-4) \\ &= 3 - 12 \\ &= -9 \end{aligned}$$

$$\begin{aligned} c_{12} &= (-1)(2) + 3(1) \\ &= -2 + 3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} c_{21} &= 4(-3) + (-2)(-4) \\ &= -12 + 8 \\ &= -4 \end{aligned}$$

$$\begin{aligned} c_{22} &= 4(2) + (-2)(1) \\ &= 8 - 2 \\ &= 6 \end{aligned}$$

$$\begin{aligned} c_{31} &= 5(-3) + 0(-4) \\ &= -15 + 0 \\ &= -15 \end{aligned}$$

$$\begin{aligned}c32 &= 5(2) + 0(1) \\&= 10 + 0 \\&= 10\end{aligned}$$

$$AB = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}$$

Example 2.1.6 - More Matrix Products

Part (a)

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix}$$

$$\begin{aligned}c11 &= 1(-2) + 0(1) + 3(-1) = -5 \\c12 &= 1(4) + 0(0) + 3(1) = 7 \\c13 &= 1(2) + 0(0) + 3(-1) = -1\end{aligned}$$

$$\begin{aligned}c21 &= 2(-2) + (-1)(1) + (-2)(-1) = -3 \\c22 &= 2(4) + (-1)(0) + (-2)(1) = 6 \\c23 &= 2(2) + (-1)(0) + (-2)(-1) = 6\end{aligned}$$

$$\text{Product} = \begin{bmatrix} -5 & 7 & -1 \\ -3 & 6 & 6 \end{bmatrix}$$

Part (b): Multiplication by identity

$$\begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$$

Part (c): Product producing the identity

$$\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\begin{aligned}c11 &= 1(-1) + 2(1) = 1 \\c12 &= 1(2) + 2(-1) = 0 \\c21 &= 1(-1) + 1(1) = 0 \\c22 &= 1(2) + 1(-1) = 1\end{aligned}$$

$$\text{Product} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Part (d): Row matrix times column matrix

$$\begin{bmatrix} 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\begin{aligned}&= 1(2) + (-2)(-1) + (-3)(1) \\&= 2 + 2 - 3 \\&= 1\end{aligned}$$

Part (e): Column matrix times row matrix

$$\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -3 \end{bmatrix}$$

$$\begin{aligned}&= \begin{bmatrix} 2(1) & 2(-2) & 2(-3) \\ -1(1) & -1(-2) & -1(-3) \\ 1(1) & 1(-2) & 1(-3) \end{bmatrix} \\&= \begin{bmatrix} 2 & -4 & -6 \\ -1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix}\end{aligned}$$

Important lesson: AB and BA are usually not equal, and sometimes they even have different sizes.

Example 2.1.7 - Solving a Matrix Equation $Ax = 0$

Solve $Ax = 0$ for the given matrix A .

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Step 1: Write the system.

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_1 + 3x_2 - 2x_3 &= 0 \end{aligned}$$

Step 2: Row-reduce the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 2 & 3 & -2 & 0 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{aligned} R_2 &= [2, 3, -2, 0] - 2[1, -2, 1, 0] \\ &= [2, 3, -2, 0] - [2, -4, 2, 0] \\ &= [0, 7, -4, 0] \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 7 & -4 & 0 \end{array} \right]$$

$$R_2 \rightarrow (1/7)R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4/7 & 0 \end{array} \right]$$

$$R_1 \rightarrow R_1 + 2R_2$$

$$\begin{aligned} R_1 &= [1, -2, 1, 0] + 2[0, 1, -4/7, 0] \\ &= [1, -2, 1, 0] + [0, 2, -8/7, 0] \\ &= [1, 0, -1/7, 0] \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1/7 & 0 \\ 0 & 1 & -4/7 & 0 \end{array} \right]$$

Step 3: Read the equations and introduce a parameter.

$$\begin{aligned} x_1 - (1/7)x_3 &= 0 & \text{so} & \quad x_1 = (1/7)x_3 \\ x_2 - (4/7)x_3 &= 0 & \text{so} & \quad x_2 = (4/7)x_3 \end{aligned}$$

$$\text{Let } x_3 = 7t.$$

$$\begin{aligned} x_1 &= (1/7)(7t) = t \\ x_2 &= (4/7)(7t) = 4t \\ x_3 &= 7t \end{aligned}$$

$$x = \begin{bmatrix} t \\ 4t \\ 7t \end{bmatrix} = t \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}$$

Example 2.1.8 - Linear Combination of Column Vectors

Write the system as a linear combination of columns and verify a solution.

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0 \\ 4x_1 + 5x_2 + 6x_3 &= 3 \\ 7x_1 + 8x_2 + 9x_3 &= 6 \end{aligned}$$

Column form:

$$x_1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 7 \end{bmatrix} \quad \begin{bmatrix} 5 \\ 8 \end{bmatrix} \quad \begin{bmatrix} 6 \\ 9 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

Check the solution $x_1 = 1$, $x_2 = 1$, $x_3 = -1$.

$$1 \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} - \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 9 \\ 15 \end{bmatrix} - \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}$$

So the right-hand side is a linear combination of the columns of A, and the system is consistent.

2.2 Properties of Matrix Operations

This section studies algebraic rules for matrices. Some rules are like ordinary numbers, but two important differences are that matrix multiplication is usually not commutative and cancellation is not always valid.

Example 2.2.1 - Addition of More Than Two Matrices

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

Add top entries and bottom entries separately.

Top entry:

$$\begin{aligned} 1 + (-1) + 0 + 2 \\ = 0 + 0 + 2 \\ = 2 \end{aligned}$$

Bottom entry:

$$\begin{aligned} 2 + (-1) + 1 + (-3) \\ = 1 + 1 - 3 \\ = -1 \end{aligned}$$

$$\text{Result} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Example 2.2.2 - Solving a Matrix Equation

Solve $3X + A = B$ for X .

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix}$$

Step 1: Isolate X .

$$\begin{aligned} 3X + A &= B \\ 3X &= B - A \\ X &= (1/3)(B - A) \end{aligned}$$

Step 2: Compute $B - A$.

$$\begin{aligned} B - A &= \begin{bmatrix} -3 & 4 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} -3 - 1 & 4 - (-2) \\ 2 - 0 & 1 - 3 \end{bmatrix} \\ &= \begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Step 3: Multiply by $1/3$.

$$\begin{aligned} X &= (1/3) \begin{bmatrix} -4 & 6 \\ 2 & -2 \end{bmatrix} \\ X &= \begin{bmatrix} -4/3 & 2 \\ 2/3 & -2/3 \end{bmatrix} \end{aligned}$$

Example 2.2.3 - Associative Property of Matrix Multiplication

Find ABC by grouping the product in two ways.

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \quad C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Method 1: Compute $(AB)C$

$$AB = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1(1)+(-2)(3) & 1(0)+(-2)(-2) & 1(2)+(-2)(1) \\ 2(1)+(-1)(3) & 2(0)+(-1)(-2) & 2(2)+(-1)(1) \end{bmatrix}$$

$$AB = \begin{bmatrix} -5 & 4 & 0 \\ -1 & 2 & 3 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} -5 & 4 & 0 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

$$c_{11} = (-5)(-1) + 4(3) + 0(2) = 17$$

$$c_{12} = (-5)(0) + 4(1) + 0(4) = 4$$

$$c_{21} = (-1)(-1) + 2(3) + 3(2) = 13$$

$$c_{22} = (-1)(0) + 2(1) + 3(4) = 14$$

$$(AB)C = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix}$$

Method 2: Compute A(BC)

$$BC = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

$$BC = \begin{bmatrix} 3 & 8 \\ -7 & 2 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ -7 & 2 \end{bmatrix}$$

$$c_{11} = 1(3) + (-2)(-7) = 17$$

$$c_{12} = 1(8) + (-2)(2) = 4$$

$$c_{21} = 2(3) + (-1)(-7) = 13$$

$$c_{22} = 2(8) + (-1)(2) = 14$$

$$A(BC) = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix}$$

Therefore, $(AB)C = A(BC)$.

Example 2.2.4 - Matrix Multiplication Is Not Commutative

Show that AB is not equal to BA.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$$

Find AB

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1(2)+3(0) & 1(-1)+3(2) \\ 2(2)+(-1)(0) & 2(-1)+(-1)(2) \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 5 \\ 4 & -4 \end{bmatrix}$$

Find BA

$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2(1)+(-1)(2) & 2(3)+(-1)(-1) \\ 0(1)+2(2) & 0(3)+2(-1) \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 7 \\ 4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -2 \end{bmatrix}$$

$$AB \neq BA.$$

Example 2.2.5 - Cancellation Is Not Always Valid

Show that $AC = BC$ even though A is not equal to B .

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1(1)+3(-1) & 1(-2)+3(2) \\ 0(1)+1(-1) & 0(-2)+1(2) \end{bmatrix}$$

$$AC = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2(1)+4(-1) & 2(-2)+4(2) \\ 2(1)+3(-1) & 2(-2)+3(2) \end{bmatrix}$$

$$BC = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

So, $AC = BC$.

But $A \neq B$.

Therefore, cancellation is not valid unless the cancelled matrix is invertible.

Example 2.2.6 - Multiplication by an Identity Matrix

Part (a)

$$\begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 4 & 0 \\ -1 & 1 \end{bmatrix}$$

The identity matrix on the right is I_2 , so $AI_2 = A$.

Part (b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$$

The identity matrix on the left is I_3 , so $I_3A = A$.

Example 2.2.7 - Repeated Multiplication of a Square Matrix

Find A^3 .

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$$

Step 1: Compute $A^2 = AA$.

$$A^2 = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 2(2)+(-1)(3) & 2(-1)+(-1)(0) \\ 3(2)+0(3) & 3(-1)+0(0) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix}$$

Step 2: Compute $A^3 = A^2A$.

$$A^3 = \begin{bmatrix} 1 & -2 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1(2)+(-2)(3) & 1(-1)+(-2)(0) \\ 6(2)+(-3)(3) & 6(-1)+(-3)(0) \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -4 & -1 \\ 3 & -6 \end{bmatrix}$$

Example 2.2.8 - Transposes of Matrices

Part (a)

$$A = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 8 \end{bmatrix}$$

Part (b)

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Part (c)

$$C = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Because $C^T = C$, the matrix C is symmetric.

Part (d)

$$D = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 4 & -1 \end{bmatrix}$$

$$D^T = \begin{bmatrix} 0 & 1 \\ 2 & 4 \\ 1 & -1 \end{bmatrix}$$

Example 2.2.9 - Transpose of a Product

Show that $(AB)^T = B^T A^T$.

$$A = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 3 & 0 \end{bmatrix}$$

Step 1: Find AB .

$$AB = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 1 \\ 6 & -1 \\ -1 & 2 \end{bmatrix}$$

Step 2: Transpose AB.

$$(AB)^T = \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Step 3: Find B^T and A^T .

$$B^T = \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -2 \\ -2 & 3 & 1 \end{bmatrix}$$

Step 4: Multiply $B^T A^T$.

$$B^T A^T = \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -2 \\ -2 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Therefore, $(AB)^T = B^T A^T$.

Example 2.2.10 - Product of a Matrix and Its Transpose

Find AA^T and show it is symmetric.

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix}$$

Step 1: Find A^T .

$$A^T = \begin{bmatrix} 1 & 0 & -2 \\ 3 & -2 & -1 \end{bmatrix}$$

Step 2: Multiply A by A^T .

$$AA^T = \begin{bmatrix} 1 & 3 \\ 0 & -2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 3 & -2 & -1 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 10 & -6 & -5 \\ -6 & 4 & 2 \\ -5 & 2 & 5 \end{bmatrix}$$

Step 3: Check symmetry. Matching entries across the main diagonal are equal.

$$a_{12} = -6 \text{ and } a_{21} = -6$$

$$a_{13} = -5 \text{ and } a_{31} = -5$$

$$a_{23} = 2 \text{ and } a_{32} = 2$$

Therefore, AA^T is symmetric.

2.3 The Inverse of a Matrix

This section explains inverse matrices, singular matrices, how to find inverses using Gauss-Jordan elimination, inverse properties, and how to solve systems by $x = A^{-1}b$.

Example 2.3.1 - Showing that B Is the Inverse of A

$$A = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$$

To prove $B = A^{-1}$, show $AB = I$ and $BA = I$.

Step 1: Find AB

$$AB = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} (-1)(1)+2(1) & (-1)(-2)+2(-1) \\ (-1)(1)+1(1) & (-1)(-2)+1(-1) \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Step 2: Find BA

$$BA = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1(-1)+(-2)(-1) & 1(2)+(-2)(1) \\ 1(-1)+(-1)(-1) & 1(2)+(-1)(1) \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $AB = BA = I$, B is the inverse of A.

Example 2.3.2 - Finding an Inverse by Solving $AX = I$

Find the inverse of A.

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$$

Let X be the unknown inverse.

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$$

Solve $AX = I$.

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Matrix multiplication gives two systems.

First column:

$$\begin{aligned} x_{11} + 4x_{21} &= 1 \\ -x_{11} - 3x_{21} &= 0 \end{aligned}$$

Second column:

$$\begin{aligned} x_{12} + 4x_{22} &= 0 \\ -x_{12} - 3x_{22} &= 1 \end{aligned}$$

Solve first column system.

$$\begin{aligned} x_{11} + 4x_{21} &= 1 \\ -x_{11} - 3x_{21} &= 0 \end{aligned}$$

Add the equations:

$$x_{21} = 1$$

Substitute:

$$x_{11} + 4(1) = 1$$

$$x_{11} = -3$$

Solve second column system.

$$x_{12} + 4x_{22} = 0$$

$$-x_{12} - 3x_{22} = 1$$

Add the equations:

$$x_{22} = 1$$

Substitute:

$$x_{12} + 4(1) = 0$$

$$x_{12} = -4$$

$$A^{-1} = X = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$

Example 2.3.3 - Finding an Inverse by Gauss-Jordan Elimination

Find A^{-1} .

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

Start with $[A \mid I]$.

$$\begin{bmatrix} 1 & -1 & 0 & | & 1 & 0 & 0 \\ 1 & 0 & -1 & | & 0 & 1 & 0 \\ -6 & 2 & 3 & | & 0 & 0 & 1 \end{bmatrix}$$

Step 1: $R_2 \rightarrow R_2 - R_1$.

$$\begin{aligned} R_2 &= [1, 0, -1 \mid 0, 1, 0] - [1, -1, 0 \mid 1, 0, 0] \\ &= [0, 1, -1 \mid -1, 1, 0] \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ -6 & 2 & 3 & | & 0 & 0 & 1 \end{bmatrix}$$

Step 2: $R_3 \rightarrow R_3 + 6R_1$.

$$\begin{aligned} 6R_1 &= [6, -6, 0 \mid 6, 0, 0] \\ R_3 + 6R_1 &= [0, -4, 3 \mid 6, 0, 1] \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & -4 & 3 & | & 6 & 0 & 1 \end{bmatrix}$$

Step 3: $R_3 \rightarrow R_3 + 4R_2$.

$$\begin{aligned} 4R_2 &= [0, 4, -4 \mid -4, 4, 0] \\ R_3 + 4R_2 &= [0, 0, -1 \mid 2, 4, 1] \end{aligned}$$

$$\begin{bmatrix} 1 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 0 & -1 & | & 2 & 4 & 1 \end{bmatrix}$$

Step 4: $R_3 \rightarrow -R_3$.

$$\begin{bmatrix} 1 & -1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & -2 & -4 & -1 \end{bmatrix}$$

Step 5: $R_2 \rightarrow R_2 + R_3$.

$$R_2 = [0, 1, -1 \mid -1, 1, 0] + [0, 0, 1 \mid -2, -4, -1]$$

$$= [0, 1, 0 \mid -3, -3, -1]$$

$$\begin{bmatrix} 1 & -1 & 0 & \mid & 1 & 0 & 0 \\ 0 & 1 & 0 & \mid & -3 & -3 & -1 \\ 0 & 0 & 1 & \mid & -2 & -4 & -1 \end{bmatrix}$$

Step 6: $R1 \rightarrow R1 + R2$.

$$\begin{aligned} R1 &= [1, -1, 0 \mid 1, 0, 0] + [0, 1, 0 \mid -3, -3, -1] \\ &= [1, 0, 0 \mid -2, -3, -1] \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & \mid & -2 & -3 & -1 \\ 0 & 1 & 0 & \mid & -3 & -3 & -1 \\ 0 & 0 & 1 & \mid & -2 & -4 & -1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{bmatrix}$$

Example 2.3.4 - Showing a Matrix Is Singular

Show that A has no inverse.

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$$

Start with $[A \mid I]$.

$$\begin{bmatrix} 1 & 2 & 0 & \mid & 1 & 0 & 0 \\ 3 & -1 & 2 & \mid & 0 & 1 & 0 \\ -2 & 3 & -2 & \mid & 0 & 0 & 1 \end{bmatrix}$$

Step 1: $R2 \rightarrow R2 - 3R1$.

$$\begin{aligned} 3R1 &= [3, 6, 0 \mid 3, 0, 0] \\ R2 - 3R1 &= [0, -7, 2 \mid -3, 1, 0] \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 0 & \mid & 1 & 0 & 0 \\ 0 & -7 & 2 & \mid & -3 & 1 & 0 \\ -2 & 3 & -2 & \mid & 0 & 0 & 1 \end{bmatrix}$$

Step 2: $R3 \rightarrow R3 + 2R1$.

$$\begin{aligned} 2R1 &= [2, 4, 0 \mid 2, 0, 0] \\ R3 + 2R1 &= [0, 7, -2 \mid 2, 0, 1] \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 0 & \mid & 1 & 0 & 0 \\ 0 & -7 & 2 & \mid & -3 & 1 & 0 \\ 0 & 7 & -2 & \mid & 2 & 0 & 1 \end{bmatrix}$$

Step 3: $R3 \rightarrow R3 + R2$.

$$R3 + R2 = [0, 0, 0 \mid -1, 1, 1]$$

$$\begin{bmatrix} 1 & 2 & 0 & \mid & 1 & 0 & 0 \\ 0 & -7 & 2 & \mid & -3 & 1 & 0 \\ 0 & 0 & 0 & \mid & -1 & 1 & 1 \end{bmatrix}$$

The left side has a zero row, so it cannot become the identity matrix.

Therefore, A has no inverse.
A is singular.

Example 2.3.5 - Inverse Formula for 2 x 2 Matrices

Part (a): Find A^{-1}

$$A = \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$$

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, use $ad - bc$.

$$a = 3, \quad b = -1, \quad c = -2, \quad d = 2$$

$$\begin{aligned} ad - bc &= (3)(2) - (-1)(-2) \\ &= 6 - 2 \\ &= 4 \end{aligned}$$

Because the determinant is not zero, the inverse exists.

$$\begin{aligned} A^{-1} &= (1/4) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= (1/4) \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 1/4 \\ 1/2 & 3/4 \end{bmatrix} \end{aligned}$$

Part (b): Check whether B^{-1} exists

$$B = \begin{bmatrix} 3 & -1 \\ -6 & 2 \end{bmatrix}$$

$$\begin{aligned} ad - bc &= (3)(2) - (-1)(-6) \\ &= 6 - 6 \\ &= 0 \end{aligned}$$

Because $ad - bc = 0$, B^{-1} does not exist.
 B is singular.

Example 2.3.6 - Computing A^{-2} in Two Ways

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

Method 1: Compute $(A^2)^{-1}$

$$A^2 = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1(1)+1(2) & 1(1)+1(4) \\ 2(1)+4(2) & 2(1)+4(4) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & 5 \\ 10 & 18 \end{bmatrix}$$

$$\begin{aligned} \det(A^2) &= 3(18) - 5(10) \\ &= 54 - 50 \\ &= 4 \end{aligned}$$

$$(A^2)^{-1} = (1/4) \begin{bmatrix} 18 & -5 \\ -10 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 9/2 & -5/4 \\ -5/2 & 3/4 \end{bmatrix}$$

Method 2: Compute $(A^{-1})^2$

$$\begin{aligned} \det(A) &= 1(4) - 1(2) \\ &= 2 \end{aligned}$$

$$A^{-1} = (1/2) \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2 & -1/2 \\ -1 & 1/2 \end{bmatrix}$$

$$(A^{-1})^2 = \begin{bmatrix} 2 & -1/2 \\ -1 & 1/2 \end{bmatrix} \begin{bmatrix} 2 & -1/2 \\ -1 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1/2 \end{bmatrix} \begin{bmatrix} -1 & 1/2 \end{bmatrix}$$

$$\begin{aligned} c_{11} &= 2(2) + (-1/2)(-1) = 9/2 \\ c_{12} &= 2(-1/2) + (-1/2)(1/2) = -5/4 \\ c_{21} &= (-1)(2) + (1/2)(-1) = -5/2 \\ c_{22} &= (-1)(-1/2) + (1/2)(1/2) = 3/4 \end{aligned}$$

$$(A^{-1})^2 = \begin{bmatrix} 9/2 & -5/4 \\ -5/2 & 3/4 \end{bmatrix}$$

Both methods give the same result, so $A^{-2} = (A^2)^{-1} = (A^{-1})^2$.

Example 2.3.7 - Inverse of a Matrix Product

Find $(AB)^{-1}$ using $(AB)^{-1} = B^{-1}A^{-1}$.

$$A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad B^{-1} = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ 2/3 & 0 & -1/3 \end{bmatrix}$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$= \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ 2/3 & 0 & -1/3 \end{bmatrix} \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Row 1:

$$\begin{aligned} c_{11} &= 1(7) + (-2)(-1) + 1(-1) = 8 \\ c_{12} &= 1(-3) + (-2)(1) + 1(0) = -5 \\ c_{13} &= 1(-3) + (-2)(0) + 1(1) = -2 \end{aligned}$$

Row 2:

$$\begin{aligned} c_{21} &= (-1)(7) + 1(-1) + 0(-1) = -8 \\ c_{22} &= (-1)(-3) + 1(1) + 0(0) = 4 \\ c_{23} &= (-1)(-3) + 1(0) + 0(1) = 3 \end{aligned}$$

Row 3:

$$\begin{aligned} c_{31} &= (2/3)(7) + 0(-1) + (-1/3)(-1) = 5 \\ c_{32} &= (2/3)(-3) + 0(1) + (-1/3)(0) = -2 \\ c_{33} &= (2/3)(-3) + 0(0) + (-1/3)(1) = -7/3 \end{aligned}$$

$$(AB)^{-1} = \begin{bmatrix} 8 & -5 & -2 \\ -8 & 4 & 3 \\ 5 & -2 & -7/3 \end{bmatrix}$$

Example 2.3.8 - Solving Systems Using an Inverse Matrix

All three systems use the same coefficient matrix A and inverse A^{-1} .

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix}$$

Part (a)

$$b = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

$$x = A^{-1}b$$

$$x = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 6 & -2 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

$$\begin{aligned} x_1 &= (-1)(-1) + 1(1) + 0(-2) = 2 \\ x_2 &= (-1)(-1) + 0(1) + 1(-2) = -1 \\ x_3 &= 6(-1) + (-2)(1) + (-3)(-2) = -2 \end{aligned}$$

$$x = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

Part (b)

$$b = \begin{bmatrix} 4 \\ 8 \\ 5 \end{bmatrix}$$

$$x_1 = (-1)(4) + 1(8) + 0(5) = 4$$

$$x_2 = (-1)(4) + 0(8) + 1(5) = 1$$

$$x_3 = 6(4) + (-2)(8) + (-3)(5) = -7$$

$$x = \begin{bmatrix} 4 \\ 1 \\ -7 \end{bmatrix}$$

Part (c)

$$b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x = A^{-1}b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This is the trivial solution.

2.4 Elementary Matrices

This section connects row operations with matrix multiplication. An elementary matrix is obtained from an identity matrix by exactly one elementary row operation.

Example 2.4.1 - Elementary and Nonelementary Matrices

Part (a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This comes from I_3 by multiplying row 2 by 3. It is elementary.

Part (b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

This is 2×3 , not square. It is not elementary.

Part (c)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This would require multiplying a row by 0, which is not allowed. It is not elementary.

Part (d)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

This comes from I_3 by interchanging rows 2 and 3. It is elementary.

Part (e)

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

This comes from I_2 by $R_2 \rightarrow R_2 + 2R_1$. It is elementary.

Part (f)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

This requires two operations: multiply row 2 by 2 and row 3 by -1. It is not elementary because an elementary matrix uses exactly one row operation.

Example 2.4.2 - Elementary Matrices Perform Row Operations

Part (a): Interchange rows

$$E = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -3 & 6 \\ 3 & 2 & -1 \end{bmatrix}$$

Because E is I_3 with rows 1 and 2 interchanged, EA interchanges rows 1 and 2 of A .

$$EA = \begin{bmatrix} 1 & -3 & 6 \\ 0 & 2 & 1 \\ 3 & 2 & -1 \end{bmatrix}$$

Part (b): Multiply one row by a constant

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & -4 & 1 \\ 0 & 2 & 6 & -4 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

E multiplies row 2 of A by 1/2.

Row 2 becomes:

$$(1/2)[0, 2, 6, -4] = [0, 1, 3, -2]$$

$$EA = \begin{bmatrix} 1 & 0 & -4 & 1 \\ 0 & 1 & 3 & -2 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

Part (c): Add a multiple of one row to another row

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$$

E performs $R_2 \rightarrow R_2 + 2R_1$ on A.

$$\begin{aligned} R_2 + 2R_1 &= [-2, -2, 3] + 2[1, 0, -1] \\ &= [-2, -2, 3] + [2, 0, -2] \\ &= [0, -2, 1] \end{aligned}$$

$$EA = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -2 & 1 \\ 0 & 4 & 5 \end{bmatrix}$$

Example 2.4.3 - Using Elementary Matrices to Reach Row-Echelon Form

Find elementary matrices that transform A into row-echelon form.

$$A = \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

Step 1: Interchange rows 1 and 2.

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1A = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

Step 2: Eliminate the 2 below the first pivot using $R_3 \rightarrow R_3 - 2R_1$.

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} R_3 - 2R_1 &= [2, -6, 2, 0] - 2[1, -3, 0, 2] \\ &= [2, -6, 2, 0] - [2, -6, 0, 4] \\ &= [0, 0, 2, -4] \end{aligned}$$

$$E_2E_1A = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & -4 \end{bmatrix}$$

Step 3: Make the third pivot 1 using $R_3 \rightarrow (1/2)R_3$.

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$$

$$E_3 E_2 E_1 A = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Therefore, the row-echelon form is $B = E_3 E_2 E_1 A$.

Example 2.4.4 - Writing a Matrix as a Product of Elementary Matrices

Write A as a product of elementary matrices.

$$A = \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}$$

Reduce A to I , then reverse the operations.

Step 1: $R_1 \rightarrow -R_1$

$$E_1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A \text{ becomes } \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$$

Step 2: $R_2 \rightarrow R_2 - 3R_1$

$$E_2 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

$$\text{Matrix becomes } \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

Step 3: $R_2 \rightarrow (1/2)R_2$

$$E_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

$$\text{Matrix becomes } \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Step 4: $R_1 \rightarrow R_1 - 2R_2$

$$E_4 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

Matrix becomes I .

So $E_4 E_3 E_2 E_1 A = I$. Therefore:

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1}$$

Find each inverse by undoing the row operation.

$$E_1^{-1} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$E_2^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

$$E_3^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$E_4^{-1} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -2 \\ 3 & 8 \end{bmatrix}$$

Example 2.4.5 - Verifying an LU-Factorization

Part (a)

$$\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$$

$$LU = \begin{bmatrix} 1(1)+0(0) & 1(2)+0(-2) \\ 1(1)+1(0) & 1(2)+1(-2) \end{bmatrix}$$

$$LU = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

Part (b)

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

$$LU = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}$$

So the given L and U multiply to the original matrix A.

Example 2.4.6 - Finding an LU-Factorization

$$A = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{bmatrix}$$

Use row operations to reduce A to upper triangular form.

Step 1: $R_3 \rightarrow R_3 - 2R_1$

$$\begin{aligned} R_3 - 2R_1 &= [2, -10, 2] - 2[1, -3, 0] \\ &= [2, -10, 2] - [2, -6, 0] \\ &= [0, -4, 2] \end{aligned}$$

Matrix becomes:

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & -4 & 2 \end{bmatrix}$$

Step 2: $R_3 \rightarrow R_3 + 4R_2$

$$\begin{aligned} R_3 + 4R_2 &= [0, -4, 2] + 4[0, 1, 3] \\ &= [0, -4, 2] + [0, 4, 12] \\ &= [0, 0, 14] \end{aligned}$$

$$U = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

The elementary matrices used in elimination are:

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$$

Because $E_2E_1A = U$, we have $A = E_1^{-1}E_2^{-1}U$. So $L = E_1^{-1}E_2^{-1}$.

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix}$$

$$L = E_1^{-1}E_2^{-1}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix}$$

$$A = LU$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix}$$

Example 2.4.7 - Solving a System Using LU-Factorization

Solve the system.

$$\begin{aligned} x_1 - 3x_2 &= -5 \\ x_2 + 3x_3 &= -1 \\ 2x_1 - 10x_2 + 2x_3 &= -20 \end{aligned}$$

Use the LU-factorization from the previous example.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} \quad b = \begin{bmatrix} -5 \\ -1 \\ -20 \end{bmatrix}$$

Step 1: Let $y = Ux$, so solve $Ly = b$ first.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -20 \end{bmatrix}$$

Equations:

$$\begin{aligned} y_1 &= -5 \\ y_2 &= -1 \\ 2y_1 - 4y_2 + y_3 &= -20 \end{aligned}$$

Substitute $y_1 = -5$ and $y_2 = -1$:

$$\begin{aligned} 2(-5) - 4(-1) + y_3 &= -20 \\ -10 + 4 + y_3 &= -20 \\ -6 + y_3 &= -20 \\ y_3 &= -14 \end{aligned}$$

So $y = [-5, -1, -14]^T$.

Step 2: Solve $Ux = y$.

$$\begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ -14 \end{bmatrix}$$

Equations:

$$\begin{aligned} x_1 - 3x_2 &= -5 \\ x_2 + 3x_3 &= -1 \\ 14x_3 &= -14 \end{aligned}$$

From $14x_3 = -14$:

$$x_3 = -1$$

Use $x_2 + 3x_3 = -1$:

$$\begin{aligned} x_2 + 3(-1) &= -1 \\ x_2 - 3 &= -1 \\ x_2 &= 2 \end{aligned}$$

Use $x_1 - 3x_2 = -5$:

$$\begin{aligned} x_1 - 3(2) &= -5 \\ x_1 - 6 &= -5 \\ x_1 &= 1 \end{aligned}$$

Solution:

$$x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

2.5 Markov Chains

This section uses stochastic matrices to model systems that move between states over time. A state matrix gives proportions in each state, and $X_{\{n\}} = P^n X_0$ gives the n th state.

Example 2.5.1 - Stochastic and Nonstochastic Matrices

Part (a)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Column sums:

$$1 + 0 + 0 = 1$$

$$0 + 1 + 0 = 1$$

$$0 + 0 + 1 = 1$$

All entries are between 0 and 1.
Therefore, A is stochastic.

Part (b)

$$B = \begin{bmatrix} 1/4 & 1/5 & 1/3 & 1/2 \\ 1/4 & 13/60 & 0 & 1/6 \\ 1/4 & 1/3 & 1/3 & 1/6 \\ 1/4 & 1/4 & 1/3 & 1/6 \end{bmatrix}$$

$$\text{Column 1: } 1/4 + 1/4 + 1/4 + 1/4 = 1$$

Column 2:

$$\begin{aligned} &1/5 + 13/60 + 1/3 + 1/4 \\ &= 12/60 + 13/60 + 20/60 + 15/60 \\ &= 60/60 \\ &= 1 \end{aligned}$$

$$\text{Column 3: } 1/3 + 0 + 1/3 + 1/3 = 1$$

$$\text{Column 4: } 1/2 + 1/6 + 1/6 + 1/6 = 1$$

Therefore, B is stochastic.

Part (c)

$$C = \begin{bmatrix} 0.9 & 0.8 \\ 0.1 & 0.2 \end{bmatrix}$$

$$\text{Column 1: } 0.9 + 0.1 = 1$$

$$\text{Column 2: } 0.8 + 0.2 = 1$$

Therefore, C is stochastic.

Part (d)

$$D = \begin{bmatrix} 1/2 & 1/4 \\ 0 & 3/4 \end{bmatrix}$$

$$\text{Column 1: } 1/2 + 0 = 1/2$$

$1/2 \neq 1$, so D is not stochastic.

Part (e)

$$E = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/4 & 3/4 & 0 \end{bmatrix}$$

Column 1:
 $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$
 $= \frac{6}{12} + \frac{4}{12} + \frac{3}{12}$
 $= \frac{13}{12}$

$\frac{13}{12} \neq 1$, so E is not stochastic.

Part (f)

$F = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \\ 0.2 & 0.3 & 0.4 & 0.5 \\ 0.3 & 0.4 & 0.5 & 0.6 \\ 0.4 & 0.5 & 0.6 & 0.7 \end{bmatrix}$

Column 2:
 $0.2 + 0.3 + 0.4 + 0.5 = 1.4$

$1.4 \neq 1$, so F is not stochastic.

Example 2.5.2 - Consumer Preference Model

The states are Company A, Company B, and no satellite service. Find the state after one year.

$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix}$ $X_0 = \begin{bmatrix} 0.1500 \\ 0.2000 \\ 0.6500 \end{bmatrix}$

$X_1 = PX_0$

First entry:
 $0.70(0.1500) + 0.15(0.2000) + 0.15(0.6500)$
 $= 0.1050 + 0.0300 + 0.0975$
 $= 0.2325$

Second entry:
 $0.20(0.1500) + 0.80(0.2000) + 0.15(0.6500)$
 $= 0.0300 + 0.1600 + 0.0975$
 $= 0.2875$

Third entry:
 $0.10(0.1500) + 0.05(0.2000) + 0.70(0.6500)$
 $= 0.0150 + 0.0100 + 0.4550$
 $= 0.4800$

$X_1 = \begin{bmatrix} 0.2325 \\ 0.2875 \\ 0.4800 \end{bmatrix}$

If there are 100,000 households, multiply each entry by 100,000.

Company A: $0.2325(100000) = 23250$
 Company B: $0.2875(100000) = 28750$
 No satellite: $0.4800(100000) = 48000$

Example 2.5.3 - Consumer Preference Over Time

Use $X_n = P^n X_0$ to estimate the number of subscribers after 3, 5, 10, and 15 years.

We already have X_1 . Now compute the next states by repeated multiplication.

$X_1 = \begin{bmatrix} 0.2325 \\ 0.2875 \\ 0.4800 \end{bmatrix}$

Find X_2

$X_2 = PX_1$

First entry:

$$\begin{aligned}
& 0.70(0.2325) + 0.15(0.2875) + 0.15(0.4800) \\
& = 0.16275 + 0.043125 + 0.07200 \\
& = 0.277875
\end{aligned}$$

Second entry:

$$\begin{aligned}
& 0.20(0.2325) + 0.80(0.2875) + 0.15(0.4800) \\
& = 0.04650 + 0.23000 + 0.07200 \\
& = 0.348500
\end{aligned}$$

Third entry:

$$\begin{aligned}
& 0.10(0.2325) + 0.05(0.2875) + 0.70(0.4800) \\
& = 0.02325 + 0.014375 + 0.33600 \\
& = 0.373625
\end{aligned}$$

$$\begin{aligned}
X_2 = & \begin{bmatrix} 0.277875 \\ 0.348500 \\ 0.373625 \end{bmatrix}
\end{aligned}$$

Find X_3

$$X_3 = PX_2$$

First entry:

$$\begin{aligned}
& 0.70(0.277875) + 0.15(0.348500) + 0.15(0.373625) \\
& = 0.30283125
\end{aligned}$$

Second entry:

$$\begin{aligned}
& 0.20(0.277875) + 0.80(0.348500) + 0.15(0.373625) \\
& = 0.39041875
\end{aligned}$$

Third entry:

$$\begin{aligned}
& 0.10(0.277875) + 0.05(0.348500) + 0.70(0.373625) \\
& = 0.306750
\end{aligned}$$

$$\begin{aligned}
X_3 \text{ approximately } & \begin{bmatrix} 0.3028 \\ 0.3904 \\ 0.3068 \end{bmatrix}
\end{aligned}$$

Interpret for 100,000 households:

$$\text{Company A after 3 years: } 0.3028(100000) = 30280$$

$$\text{Company B after 3 years: } 0.3904(100000) = 39040$$

Later years

$$\begin{aligned}
X_5 \text{ approximately } & \begin{bmatrix} 0.3241 \\ 0.4381 \\ 0.2378 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
X_{10} \text{ approximately } & \begin{bmatrix} 0.3329 \\ 0.4715 \\ 0.1957 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
X_{15} \text{ approximately } & \begin{bmatrix} 0.3333 \\ 0.4756 \\ 0.1911 \end{bmatrix}
\end{aligned}$$

Example 2.5.4 - Regular Stochastic Matrices

Part (a)

$$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix}$$

Every entry is positive, so P itself already has only positive entries. Therefore, P is regular.

Part (b)

$$P = \begin{bmatrix} 0.50 & 1.00 \\ 0.50 & 0 \end{bmatrix}$$

P has a zero entry, so check P^2 .

$$P^2 = \begin{bmatrix} 0.50 & 1.00 \\ 0.50 & 0 \end{bmatrix} \begin{bmatrix} 0.50 & 1.00 \\ 0.50 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.50(0.50)+1.00(0.50) & 0.50(1.00)+1.00(0) \\ 0.50(0.50)+0(0.50) & 0.50(1.00)+0(0) \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.75 & 0.50 \\ 0.25 & 0.50 \end{bmatrix}$$

All entries of P^2 are positive. Therefore, P is regular.

Part (c)

$$P = \begin{bmatrix} 1/3 & 0 & 1 \\ 1/3 & 1 & 0 \\ 1/3 & 0 & 0 \end{bmatrix}$$

The second column is $[0, 1, 0]^T$, meaning state 2 stays in state 2 forever. Powers of P keep zeros in that column. Therefore, no power of P has all positive entries, so P is not regular.

Example 2.5.5 - Finding a Steady State Matrix

Find the steady state X for the satellite TV transition matrix P.

$$P = \begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.20 & 0.80 & 0.15 \\ 0.10 & 0.05 & 0.70 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

A steady state satisfies $PX = X$ and $x_1 + x_2 + x_3 = 1$.

$PX = X$ gives:

$$0.70x_1 + 0.15x_2 + 0.15x_3 = x_1$$

$$0.20x_1 + 0.80x_2 + 0.15x_3 = x_2$$

$$0.10x_1 + 0.05x_2 + 0.70x_3 = x_3$$

Move terms to the left.

$$-0.30x_1 + 0.15x_2 + 0.15x_3 = 0$$

$$0.20x_1 - 0.20x_2 + 0.15x_3 = 0$$

$$0.10x_1 + 0.05x_2 - 0.30x_3 = 0$$

Use the first equation.

$$-0.30x_1 + 0.15x_2 + 0.15x_3 = 0$$

Multiply by 20:

$$-6x_1 + 3x_2 + 3x_3 = 0$$

Divide by 3:

$$-2x_1 + x_2 + x_3 = 0$$

$$x_2 + x_3 = 2x_1$$

$$x_2 = 2x_1 - x_3$$

Use the second equation.

$$0.20x_1 - 0.20x_2 + 0.15x_3 = 0$$

Multiply by 20:

$$4x_1 - 4x_2 + 3x_3 = 0$$

$$\begin{aligned}
&\text{Substitute } x_2 = 2x_1 - x_3: \\
&4x_1 - 4(2x_1 - x_3) + 3x_3 = 0 \\
&4x_1 - 8x_1 + 4x_3 + 3x_3 = 0 \\
&-4x_1 + 7x_3 = 0 \\
&7x_3 = 4x_1 \\
&x_3 = (4/7)x_1
\end{aligned}$$

Find x_2 .

$$\begin{aligned}
x_2 &= 2x_1 - x_3 \\
x_2 &= 2x_1 - (4/7)x_1 \\
x_2 &= (14/7)x_1 - (4/7)x_1 \\
x_2 &= (10/7)x_1
\end{aligned}$$

Use $x_1 + x_2 + x_3 = 1$.

$$\begin{aligned}
x_1 + (10/7)x_1 + (4/7)x_1 &= 1 \\
(7/7)x_1 + (10/7)x_1 + (4/7)x_1 &= 1 \\
(21/7)x_1 &= 1 \\
3x_1 &= 1 \\
x_1 &= 1/3 \\
x_2 &= (10/7)(1/3) = 10/21 \\
x_3 &= (4/7)(1/3) = 4/21
\end{aligned}$$

$$X = \begin{bmatrix} 1/3 \\ 10/21 \\ 4/21 \end{bmatrix}$$

Example 2.5.6 - Absorbing and Nonabsorbing Markov Chains

Part (a)

$$P = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 1 & 0.5 \\ 0.6 & 0 & 0.5 \end{bmatrix}$$

Column 2 is $[0, 1, 0]^T$, so state S_2 is absorbing. From S_3 there is probability 0.5 of going to S_2 . From S_1 , one can go to S_3 and then to S_2 . Therefore, the chain is absorbing.

Part (b)

$$P = \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0.5 & 1 & 0 & 0 \\ 0 & 0 & 0.4 & 0.5 \\ 0 & 0 & 0.6 & 0.5 \end{bmatrix}$$

State S_2 is absorbing, but S_3 and S_4 only move between each other and cannot reach S_2 . Therefore, the chain is not absorbing.

Part (c)

$$P = \begin{bmatrix} 0.5 & 0 & 0.2 & 0 \\ 0.2 & 1 & 0.3 & 0 \\ 0.1 & 0 & 0.4 & 0 \\ 0.2 & 0 & 0.1 & 1 \end{bmatrix}$$

Column 2 is $[0, 1, 0, 0]^T$, so S_2 is absorbing. Column 4 is $[0, 0, 0, 1]^T$, so S_4 is absorbing. States S_1 and S_3 can reach an absorbing state. Therefore, the chain is absorbing.

Example 2.5.7 - Steady State Matrices of Absorbing Markov Chains

Part (a)

$$P = \begin{bmatrix} 0.4 & 0 & 0 \\ 0 & 1 & 0.5 \\ 0.6 & 0 & 0.5 \end{bmatrix}$$

Let $X = [x_1, x_2, x_3]^T$. Solve $PX = X$ and $x_1 + x_2 + x_3 = 1$.

$$\begin{aligned} 0.4x_1 &= x_1 \\ x_2 + 0.5x_3 &= x_2 \\ 0.6x_1 + 0.5x_3 &= x_3 \end{aligned}$$

$$\begin{aligned} \text{From } 0.4x_1 &= x_1: \\ -0.6x_1 &= 0 \\ x_1 &= 0 \end{aligned}$$

$$\begin{aligned} \text{From } x_2 + 0.5x_3 &= x_2: \\ 0.5x_3 &= 0 \\ x_3 &= 0 \end{aligned}$$

$$\begin{aligned} \text{Using } x_1 + x_2 + x_3 &= 1: \\ 0 + x_2 + 0 &= 1 \\ x_2 &= 1 \end{aligned}$$

$$X = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Part (b)

$$P = \begin{bmatrix} 0.5 & 0 & 0.2 & 0 \\ 0.2 & 1 & 0.3 & 0 \\ 0.1 & 0 & 0.4 & 0 \\ 0.2 & 0 & 0.1 & 1 \end{bmatrix}$$

Let $X = [x_1, x_2, x_3, x_4]^T$. Solving $PX = X$ gives $x_1 = 0$ and $x_3 = 0$. Then the sum equation gives $x_2 + x_4 = 1$.

$$\begin{aligned} \text{Let } x_4 &= t. \\ \text{Then } x_2 &= 1 - t. \end{aligned}$$

$$X = \begin{bmatrix} 0 \\ 1 - t \\ 0 \\ t \end{bmatrix}$$

$$0 \leq t \leq 1$$

There are infinitely many steady states because there are two absorbing states.

2.6 More Applications of Matrix Operations

This section applies matrices to cryptography, Leontief input-output models, and least squares regression.

Example 2.6.1 - Forming Uncoded Row Matrices

Write the uncoded row matrices of size 1×3 for the message MEET ME MONDAY.

Use the code 0 = blank, 1 = A, 2 = B, ..., 26 = Z.

M = 13
E = 5
E = 5
T = 20
blank = 0
M = 13
E = 5
blank = 0
M = 13
O = 15
N = 14
D = 4
A = 1
Y = 25

Group the numbers in threes. Add a blank 0 at the end to complete the last row.

[13 5 5]
[20 0 13]
[5 0 13]
[15 14 4]
[1 25 0]

Example 2.6.2 - Encoding a Message

Use the encoding matrix A to encode MEET ME MONDAY.

A = [1 -2 2]
 [-1 1 3]
 [1 -1 -4]

Encoding rule: $Y = XA$, where X is an uncoded row matrix.

First row

[13 5 5]A

First entry: $13(1) + 5(-1) + 5(1) = 13$

Second entry: $13(-2) + 5(1) + 5(-1) = -26$

Third entry: $13(2) + 5(3) + 5(-4) = 21$

[13 5 5]A = [13 -26 21]

Second row

[20 0 13]A

First entry: $20(1) + 0(-1) + 13(1) = 33$

Second entry: $20(-2) + 0(1) + 13(-1) = -53$

Third entry: $20(2) + 0(3) + 13(-4) = -12$

[20 0 13]A = [33 -53 -12]

Third row

[5 0 13]A

First entry: $5(1) + 0(-1) + 13(1) = 18$
 Second entry: $5(-2) + 0(1) + 13(-1) = -23$
 Third entry: $5(2) + 0(3) + 13(-4) = -42$

$$[5 \ 0 \ 13]A = [18 \ -23 \ -42]$$

Fourth row

$$[15 \ 14 \ 4]A$$

First entry: $15(1) + 14(-1) + 4(1) = 5$
 Second entry: $15(-2) + 14(1) + 4(-1) = -20$
 Third entry: $15(2) + 14(3) + 4(-4) = 56$

$$[15 \ 14 \ 4]A = [5 \ -20 \ 56]$$

Fifth row

$$[1 \ 25 \ 0]A$$

First entry: $1(1) + 25(-1) + 0(1) = -24$
 Second entry: $1(-2) + 25(1) + 0(-1) = 23$
 Third entry: $1(2) + 25(3) + 0(-4) = 77$

$$[1 \ 25 \ 0]A = [-24 \ 23 \ 77]$$

Encoded cryptogram:

13, -26, 21, 33, -53, -12, 18, -23, -42, 5, -20, 56, -24, 23, 77

Example 2.6.3 - Decoding a Message

Decode the cryptogram using A^{-1} .

$$A^{-1} = \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix}$$

Group the cryptogram into rows of 3.

$$\begin{bmatrix} 13 & -26 & 21 \\ 33 & -53 & -12 \\ 18 & -23 & -42 \\ 5 & -20 & 56 \\ -24 & 23 & 77 \end{bmatrix}$$

Decode row 1

$$[13 \ -26 \ 21]A^{-1}$$

First entry: $13(-1) + (-26)(-1) + 21(0) = 13$
 Second entry: $13(-10) + (-26)(-6) + 21(-1) = 5$
 Third entry: $13(-8) + (-26)(-5) + 21(-1) = 5$

Decoded row: $[13 \ 5 \ 5] = M \ E \ E$

Decode row 2

$$[33 \ -53 \ -12]A^{-1}$$

First entry: $33(-1) + (-53)(-1) + (-12)(0) = 20$
 Second entry: $33(-10) + (-53)(-6) + (-12)(-1) = 0$
 Third entry: $33(-8) + (-53)(-5) + (-12)(-1) = 13$

Decoded row: $[20 \ 0 \ 13] = T \ \text{blank} \ M$

Decode row 3

$$[18 \ -23 \ -42]A^{-1}$$

First entry: $18(-1) + (-23)(-1) + (-42)(0) = 5$
 Second entry: $18(-10) + (-23)(-6) + (-42)(-1) = 0$
 Third entry: $18(-8) + (-23)(-5) + (-42)(-1) = 13$

Decoded row: $[5 \ 0 \ 13] = \text{E blank M}$

Decode row 4

$[5 \ -20 \ 56]A^{-1}$

First entry: $5(-1) + (-20)(-1) + 56(0) = 15$
 Second entry: $5(-10) + (-20)(-6) + 56(-1) = 14$
 Third entry: $5(-8) + (-20)(-5) + 56(-1) = 4$

Decoded row: $[15 \ 14 \ 4] = \text{O N D}$

Decode row 5

$[-24 \ 23 \ 77]A^{-1}$

First entry: $(-24)(-1) + 23(-1) + 77(0) = 1$
 Second entry: $(-24)(-10) + 23(-6) + 77(-1) = 25$
 Third entry: $(-24)(-8) + 23(-5) + 77(-1) = 0$

Decoded row: $[1 \ 25 \ 0] = \text{A Y blank}$

Decoded message: MEET ME MONDAY

Example 2.6.4 - Forming an Input-Output Matrix

A simple economy has electricity E, water W, and coal C. The columns describe what each industry needs to produce one unit of output.

Electricity production needs:
 0.5 electricity, 0.25 water, 0.25 coal

Water production needs:
 0.1 electricity, 0.6 water, 0 coal

Coal production needs:
 0.2 electricity, 0.15 water, 0.5 coal

Build the columns.

Electricity column = $\begin{bmatrix} 0.5 \\ 0.25 \\ 0.25 \end{bmatrix}$

Water column = $\begin{bmatrix} 0.1 \\ 0.6 \\ 0 \end{bmatrix}$

Coal column = $\begin{bmatrix} 0.2 \\ 0.15 \\ 0.5 \end{bmatrix}$

$D = \begin{bmatrix} 0.5 & 0.1 & 0.2 \\ 0.25 & 0.6 & 0.15 \\ 0.25 & 0 & 0.5 \end{bmatrix}$

Example 2.6.5 - Solving an Open Leontief Input-Output Model

Given input-output matrix D and external demand E, find output X.

$D = \begin{bmatrix} 0.1 & 0.43 & 0 \\ 0.15 & 0 & 0.37 \\ 0.23 & 0.03 & 0.02 \end{bmatrix}$ $E = \begin{bmatrix} 20000 \\ 30000 \\ 25000 \end{bmatrix}$

Use the open-system formula.

$$\begin{aligned}X &= DX + E \\X - DX &= E \\(I - D)X &= E \\X &= (I - D)^{-1}E\end{aligned}$$

Step 1: Find $I - D$.

$$\begin{aligned}I - D &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.1 & 0.43 & 0 \\ 0.15 & 0 & 0.37 \\ 0.23 & 0.03 & 0.02 \end{bmatrix} \\I - D &= \begin{bmatrix} 0.9 & -0.43 & 0 \\ -0.15 & 1 & -0.37 \\ -0.23 & -0.03 & 0.98 \end{bmatrix}\end{aligned}$$

Step 2: Use the inverse, approximately.

$$\begin{aligned}(I - D)^{-1} &\text{approximately} \\&\begin{bmatrix} 1.25 & 0.55 & 0.21 \\ 0.30 & 1.14 & 0.43 \\ 0.30 & 0.16 & 1.08 \end{bmatrix}\end{aligned}$$

Step 3: Multiply by E .

$$\begin{aligned}X &\text{approximately} \begin{bmatrix} 1.25 & 0.55 & 0.21 \\ 0.30 & 1.14 & 0.43 \\ 0.30 & 0.16 & 1.08 \end{bmatrix} \begin{bmatrix} 20000 \\ 30000 \\ 25000 \end{bmatrix}\end{aligned}$$

First entry:

$$\begin{aligned}&1.25(20000) + 0.55(30000) + 0.21(25000) \\&= 25000 + 16500 + 5250 \\&= 46750\end{aligned}$$

Second entry:

$$\begin{aligned}&0.30(20000) + 1.14(30000) + 0.43(25000) \\&= 6000 + 34200 + 10750 \\&= 50950\end{aligned}$$

Third entry:

$$\begin{aligned}&0.30(20000) + 0.16(30000) + 1.08(25000) \\&= 6000 + 4800 + 27000 \\&= 37800\end{aligned}$$

$$\begin{aligned}X &\text{approximately} \begin{bmatrix} 46750 \\ 50950 \\ 37800 \end{bmatrix}\end{aligned}$$

Example 2.6.6 - Comparing Two Straight-Line Approximations

For the data points (1,1), (2,2), (3,4), (4,4), (5,6), compare two models by their sum of squared errors.

$$\begin{aligned}\text{Model 1: } f_1(x) &= 0.5 + x \\ \text{Model 2: } f_2(x) &= 1.2x\end{aligned}$$

Model 1

$$\begin{aligned}\text{At } x = 1: f_1(1) &= 1.5, \text{ error} = 1 - 1.5 = -0.5, \text{ square} = 0.25 \\ \text{At } x = 2: f_1(2) &= 2.5, \text{ error} = 2 - 2.5 = -0.5, \text{ square} = 0.25 \\ \text{At } x = 3: f_1(3) &= 3.5, \text{ error} = 4 - 3.5 = 0.5, \text{ square} = 0.25 \\ \text{At } x = 4: f_1(4) &= 4.5, \text{ error} = 4 - 4.5 = -0.5, \text{ square} = 0.25 \\ \text{At } x = 5: f_1(5) &= 5.5, \text{ error} = 6 - 5.5 = 0.5, \text{ square} = 0.25\end{aligned}$$

$$\begin{aligned}\text{SSE1} &= 0.25 + 0.25 + 0.25 + 0.25 + 0.25 \\&= 1.25\end{aligned}$$

Model 2

At $x = 1$: $f_2(1) = 1.2$, error = $1 - 1.2 = -0.2$, square = 0.04
 At $x = 2$: $f_2(2) = 2.4$, error = $2 - 2.4 = -0.4$, square = 0.16
 At $x = 3$: $f_2(3) = 3.6$, error = $4 - 3.6 = 0.4$, square = 0.16
 At $x = 4$: $f_2(4) = 4.8$, error = $4 - 4.8 = -0.8$, square = 0.64
 At $x = 5$: $f_2(5) = 6.0$, error = $6 - 6.0 = 0$, square = 0

$$\begin{aligned} \text{SSE}_2 &= 0.04 + 0.16 + 0.16 + 0.64 + 0 \\ &= 1.00 \end{aligned}$$

Because $1.00 < 1.25$, the model $f_2(x) = 1.2x$ fits better than $f_1(x) = 0.5 + x$.

Example 2.6.7 - Finding the Least Squares Regression Line

Find the least squares regression line for the data points $(1,1)$, $(2,2)$, $(3,4)$, $(4,4)$, $(5,6)$.

The line has the form $y = a_0 + a_1x$. Use the matrix formula $A = (X^T X)^{-1} X^T Y$.

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix}$$

Step 1: Find X^T .

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$

Step 2: Compute $X^T X$.

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}$$

Entry $(1,1)$: $1 + 1 + 1 + 1 + 1 = 5$
 Entry $(1,2)$: $1 + 2 + 3 + 4 + 5 = 15$
 Entry $(2,1)$: $1 + 2 + 3 + 4 + 5 = 15$
 Entry $(2,2)$: $1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

$$X^T X = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$$

Step 3: Compute $X^T Y$.

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 4 \\ 6 \end{bmatrix}$$

First entry:
 $1 + 2 + 4 + 4 + 6 = 17$

Second entry:
 $1(1) + 2(2) + 3(4) + 4(4) + 5(6)$
 $= 1 + 4 + 12 + 16 + 30$
 $= 63$

$$X^T Y = \begin{bmatrix} 17 \\ 63 \end{bmatrix}$$

Step 4: Find $(X^T X)^{-1}$.

$$X^T X = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$$

$$\text{determinant} = 5(55) - 15(15)$$

$$= 275 - 225$$

$$= 50$$

$$(X^T X)^{-1} = (1/50) \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix}$$

Step 5: Find the coefficient matrix A.

$$A = (X^T X)^{-1} X^T Y$$

$$A = (1/50) \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} 17 \\ 63 \end{bmatrix}$$

First entry:

$$(1/50)[55(17) + (-15)(63)]$$

$$= (1/50)[935 - 945]$$

$$= -10/50$$

$$= -0.2$$

Second entry:

$$(1/50)[(-15)(17) + 5(63)]$$

$$= (1/50)[-255 + 315]$$

$$= 60/50$$

$$= 1.2$$

$$A = \begin{bmatrix} -0.2 \\ 1.2 \end{bmatrix}$$

Therefore:

$$a_0 = -0.2$$

$$a_1 = 1.2$$

Least squares regression line:

$$y = -0.2 + 1.2x$$

Step 6: Verify with sum of squared error.

$$f(x) = -0.2 + 1.2x$$

$$x = 1: f(1) = 1.0, \text{ error} = 1 - 1.0 = 0, \text{ square} = 0$$

$$x = 2: f(2) = 2.2, \text{ error} = 2 - 2.2 = -0.2, \text{ square} = 0.04$$

$$x = 3: f(3) = 3.4, \text{ error} = 4 - 3.4 = 0.6, \text{ square} = 0.36$$

$$x = 4: f(4) = 4.6, \text{ error} = 4 - 4.6 = -0.6, \text{ square} = 0.36$$

$$x = 5: f(5) = 5.8, \text{ error} = 6 - 5.8 = 0.2, \text{ square} = 0.04$$

$$SSE = 0 + 0.04 + 0.36 + 0.36 + 0.04$$

$$= 0.80$$

This is smaller than the two earlier visual models, so the least squares line is the best of the models discussed.

Chapter 2 Formula Sheet

Matrix addition:

$$A + B = [a_{ij} + b_{ij}]$$

Scalar multiplication:

$$cA = [ca_{ij}]$$

Matrix multiplication entry:

$$(AB)_{ij} = \text{sum from } k = 1 \text{ to } n \text{ of } a_{ik} b_{kj}$$

Identity matrix:

$$AI_n = A$$

$$I_mA = A$$

Inverse matrix:

$$AA^{-1} = A^{-1}A = I$$

Gauss-Jordan inverse method:

$$[A \mid I] \rightarrow [I \mid A^{-1}]$$

2 x 2 inverse:

$$\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{then } A^{-1} = \frac{1}{(ad - bc)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

provided $ad - bc \neq 0$.

Transpose of a product:

$$(AB)^T = B^T A^T$$

Product inverse rule:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Markov chains:

$$X_n = P^n X_0$$

$$\text{Steady state: } PX = X$$

Open Leontief model:

$$X = (I - D)^{-1}E$$

Least squares regression:

$$A = (X^T X)^{-1} X^T Y$$